Lecture 7
Advanced Topics in Shear Stress

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Textbook: Gere & Goodno, Mechanics of Materials, Eighth Edition, SI Edition

Outline

- Shear stresses in thin-walled open cross sections
- Shear stresses in wide-flange beams
- Shear centers of thin-walled open sections

What you should be able to do

- Say what a thin-walled open section is, and why the shear formula of Lecture 6 is not enough for one.
- Derive the shear-stress and shear-flow formulas for such a section.
- Find the shear stresses in the flange of a wide-flange beam.
- Locate the shear center of a typical section.

The shear center

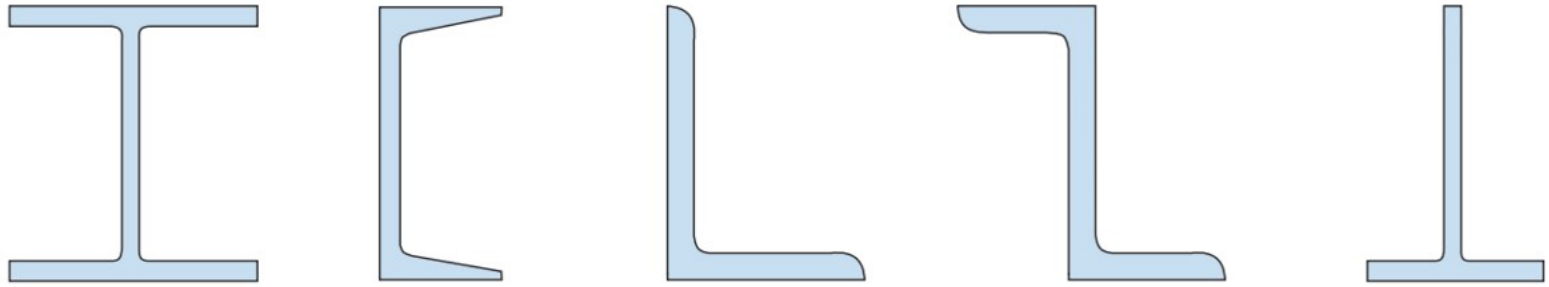
The shear center (also called the flexural center or center of twist) is the unique point on a beam's cross-section through which a transverse load must pass to cause pure bending without any twisting or torsion.

If the load misses it, the beam bends and twists as well.

Lecture 7

Thin-Walled Open Cross Sections

What a thin-walled open section is



- The wall thickness is small compared with the height and width of the section.
- The cross section is open — an I-beam or a channel — not closed like a hollow box.
- Also called structural sections or profile sections.

These shapes are weak in torsion, so it matters where the load is applied.

The shear formula from Lecture 6

$$\tau = \frac{V\,Q}{I\,b}$$

- τ is the shear stress at the level considered.
- V is the shear force on the cross section.
- I is the second moment about the neutral axis.
- b is the width at the level considered.
- Q is the first moment of the area beyond that level.

Why that is not enough

The formula assumes the stress is uniform across the width b.

In a wide flange that fails: the width jumps from the web to the full flange.

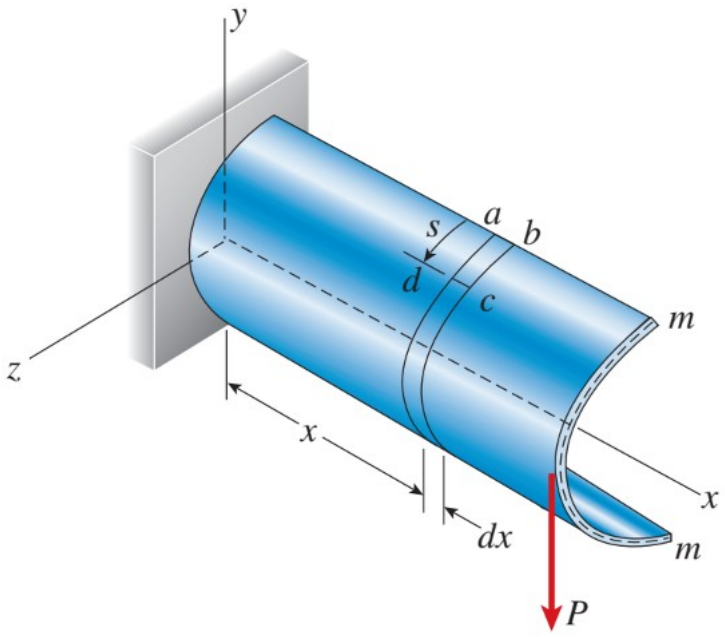
We need a formula that follows the wall, from the free edge inwards.

The general set-up

Take a beam whose cross-sectional centerline has an arbitrary shape.

The *y* and *z* axes are principal centroidal axes, and the load *P* acts parallel to the *y* axis through the shear center S.

The beam then bends in the xy plane, with the *z* axis as the neutral axis.



The normal stress

Under those conditions the normal stress σ_x at any point comes from the flexure formula:

$$\sigma_x = \frac{M_z\, y}{I_z} \tag{1}$$

- M_z is the bending moment about the z axis.
- y is the coordinate of the point, measured from the neutral axis.
- I_z is the second moment of the area about the z axis.

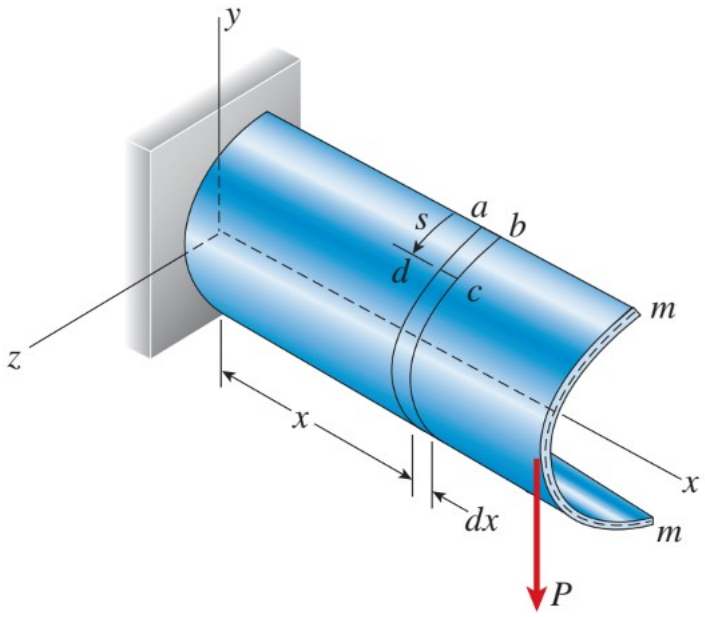
The volume element abcd

Cut a volume element abcd out between two cross sections a distance *dx* apart.

The element begins at the edge of the cross section.

Its length s is measured along the centerline, from that edge.

So s locates the cut: it runs from the free edge inwards along the wall.

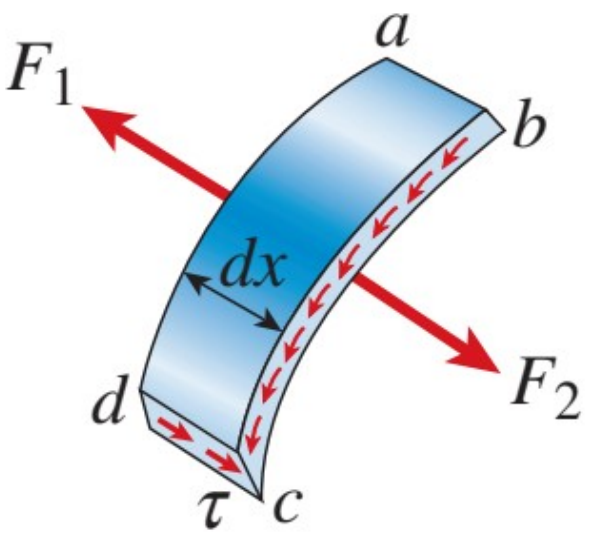


The element on its own

On face ad the normal stresses add up to the force F_1, and on face bc to the force F_2.

The bending moment is larger at ad, so F_1 is larger than F_2.

The element can only balance if a shear stress τ acts along face cd.



That τ is matched by complementary shear stresses on the cross-sectional faces ad and bc.

Horizontal equilibrium

Sum the forces in the *x* direction on the element:

$$\tau\, t\, dx + F_2 - F_1 = 0 \quad \text{so} \quad \tau\, t\, dx = F_1 - F_2 \qquad (2)$$

Here *t* is the thickness of the cross section at face cd, that is, the thickness at distance s from the free edge.

The thickness need not be constant: it may vary with s.

The force on each end face

Integrate the normal stress over the area A of the cross section, from the free edge to the cut:

$$F_1 = \int_0^s \sigma_x \, dA = \frac{M_{z1}}{I_z} \int_0^s y \, dA$$

$$F_2 = \int_0^s \sigma_x \, dA = \frac{M_{z2}}{I_z} \int_0^s y \, dA$$

M_{z1} and M_{z2} are the bending moments at the two faces.

Putting the two together

Substitute those two expressions into the equilibrium equation and solve for the shear stress:

$$\tau = \left(\frac{M_{z2} - M_{z1}}{dx}\right)\left(\frac{1}{I_z\, t}\right)\int_0^s y\, dA \qquad (3)$$

The change in moment between the two sections is what drives the shear stress.

The change in moment is the shear force

The rate of change of the bending moment is the shear force:

$$\frac{M_{z2} - M_{z1}}{dx} = \frac{dM_z}{dx} = V_y \qquad (4)$$

V_y is parallel to the y axis and positive in the direction of the load P

The shear stress at distance s

Putting $dM/dx = V_y$ back into the expression gives

$$\tau = \frac{V_y}{I_z\, t}\int_0^s y\, dA \qquad (5)$$

This is the shear stress at any point of the cross section, at distance s from the free edge.

The integral is over the area from the free edge to the cut.

The first moment, and the shear formula

The integral is the first moment of that area about the *z* axis, the neutral axis. Call it Q_z:

$$Q_z = \int_0^s y \, dA$$

With that notation the shear stress is

$$\tau = \frac{V_y \, Q_z}{I_z \, t} \qquad (6)$$

It has the same form as the shear formula, with *b* replaced by the wall thickness *t*.

Shear flow

The shear flow is the shear force per unit length along the wall, the shear stress times the thickness:

$$f = \tau t = \frac{V_y Q_z}{I_z} \tag{7}$$

- V_y and I_z are constants, so f is proportional to Q_z.
- At the top and bottom edges Q_z is zero, so the shear flow is zero there.
- It varies continuously between those edges and peaks where Q_z is largest, at the neutral axis.

Bending about the other axis

If the beam is instead bent by loads parallel to the z axis through the shear center, the y axis becomes the neutral axis.

The same analysis then gives

$$\tau = \frac{V_z\, Q_y}{I_y\, t}, \qquad f = \frac{V_z\, Q_y}{I_y} \tag{8}$$

V_z is the shear force along z, Q_y the first moment about y, and I_y the second moment about y.

Where the stresses point

- The shear stresses are directed along the centerline of the section, and act parallel to the edges.
- Across the thickness *t* the stress is taken as constant — a good assumption when the wall is thin.
- The wall thickness may vary with s.

What the derivation requires

- The shear force must act through the shear center.
- It must be parallel to one of the principal centroidal axes.
- If it is inclined to both axes, resolve it into components parallel to each, analyse the two cases separately, and superpose.

The next section puts these equations to work on a wide-flange beam.

Lecture 7

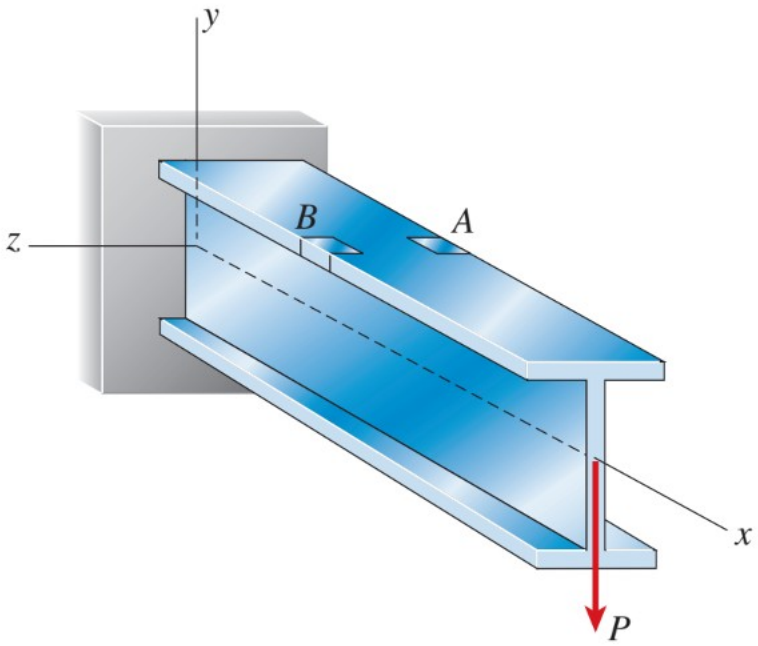
Shear Stresses in Wide-Flange Beams

The beam, and where the load acts

A wide-flange beam carries a force P in the plane of the web.

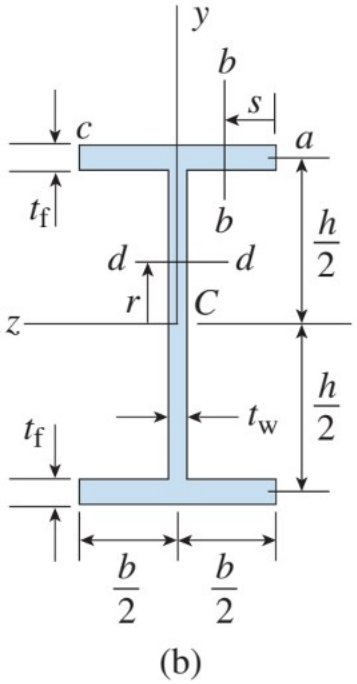
That plane contains the shear center, which for this doubly symmetric section is the centroid.

Lecture 6 found the shear stress in the web alone. Now the flanges are included as well.



The dimensions of the section

- b is the flange width.
- h is the height between the centerlines of the flanges.
- t_f is the flange thickness.
- t_w is the web thickness.



(b)

All the calculations use these centerline dimensions.

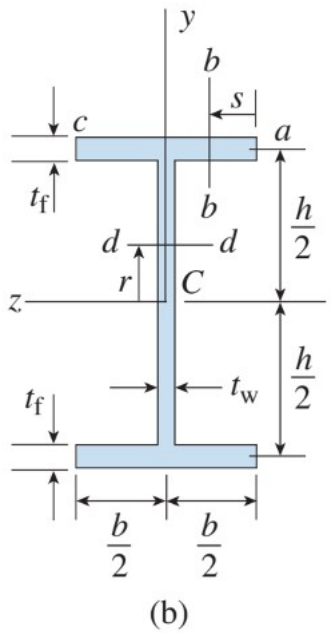
The upper flange

Cut the flange at bb, a distance s from the free edge a. The area between them is s times the flange thickness:

$$A = s\,t_f$$

Its centroid is $h/2$ from the neutral axis, so its first moment is

$$Q_z = s\,t_f\,\frac{h}{2} \qquad (9)$$



(b)

The stress in the flange

Putting that first moment into the shear formula gives the stress τ_f in the flange:

$$\tau_f = \frac{V_y\, Q_z}{I_z\, t} = \frac{P\,(s\, t_f\, h/2)}{I_z\, t_f} = \frac{s\, h\, P}{2\, I_z} \qquad (10)$$

The thickness *t* in the formula is the flange thickness, and it cancels.

The stress grows linearly with the distance s from the free edge.

Maximum in the flange

The largest value, τ_1, is at the junction with the web, where s = $b/2$:

$$\tau_1 = \frac{b\,h\,P}{4\,I_z} \tag{11}$$

The matching shear flow f_1 is that stress times the flange thickness:

$$f_1 = \tau_1\,t_f = \frac{b\,h\,t_f\,P}{4\,I_z} \tag{12}$$

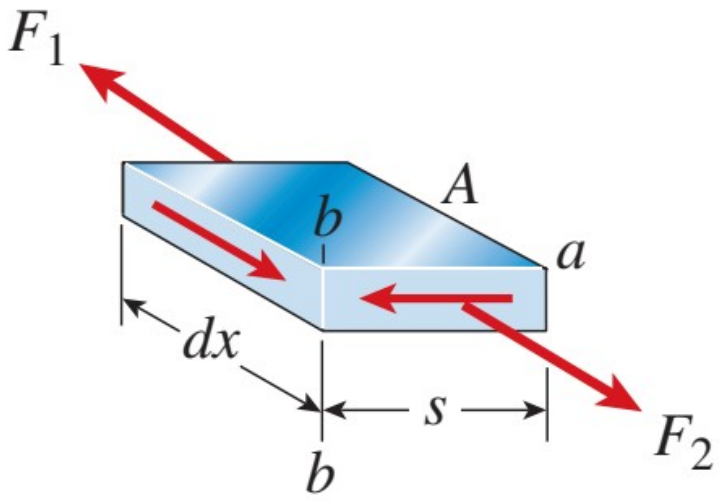
At the free edge s is zero, so the stress and the flow are zero there.

Which way does it act?

Cut element A out of the flange.

The moment is larger on its rear face, so F_1 exceeds F_2.

For equilibrium the stress on its left face acts toward the reader.



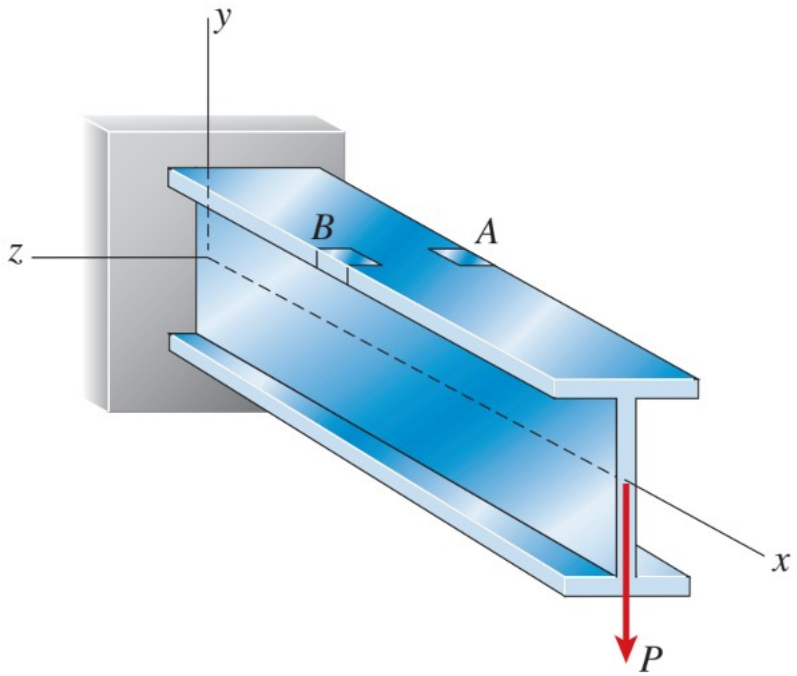
So on the cross section it acts to the left.

The other half of the flange

Measuring s from c, on the left, gives the same magnitudes.

Element B: the stresses there act to the right.

So both halves feed shear into the web.

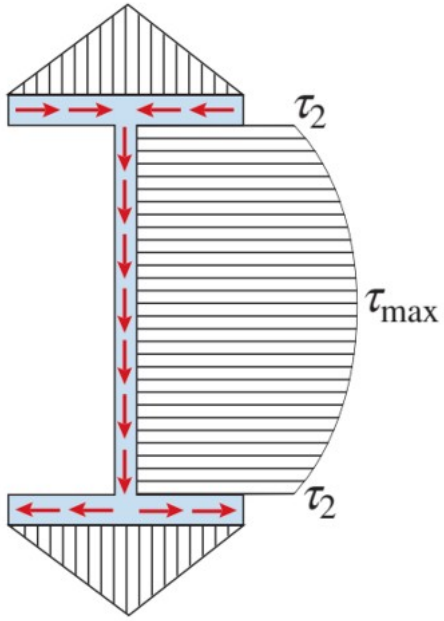


The lower flange has the same magnitudes; only the directions change.

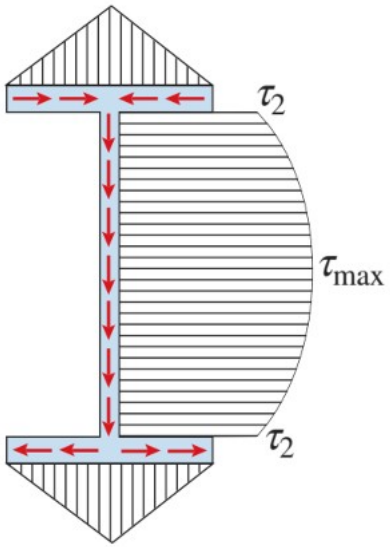
The lower flange

The same analysis, starting from the free edge of the lower flange, gives the same magnitudes.

So the section carries shear stresses in its flanges as well as its web.



The flow of shear



The stresses flow in from the outer edges of the top flange, down through the web, and out to the edges of the bottom flange.

That flow is continuous everywhere in the section, so its direction can be read off directly — much quicker than cutting out elements and balancing them.

The second moment of the section

On centerline dimensions, the second moment of the whole section is

$$I_z = \frac{t_w\, h^3}{12} + \frac{b\, t_f\, h^2}{2} \qquad (13)$$

The first term is the web; the second is the two flanges, each of area b times t_f, at distance $h/2$.

The resultant is the load

Lecture 6 showed that the web stresses alone add up to the load:

The flange stresses are horizontal and cancel in pairs, so they add nothing.

The resultant acts in the plane of the web, through the shear center.

The next section uses that property to locate other shear centers.

What this adds to the earlier analysis

- Lecture 6 found only the web stress, and only for a wide-flange beam.
- This adds the flange stress: zero at the free edge, largest at the web.
- The shear flows in from both edges of a flange and meets at the web.
- The method works for any thin-walled open section.

Lecture 7

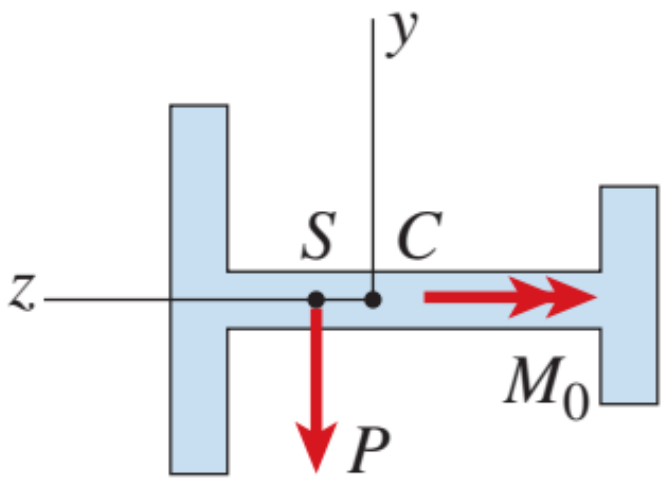
Shear centers of Thin-Walled Open Sections

Why the shear center matters

Suppose a lateral load is applied at some point *A*, not at the shear center.

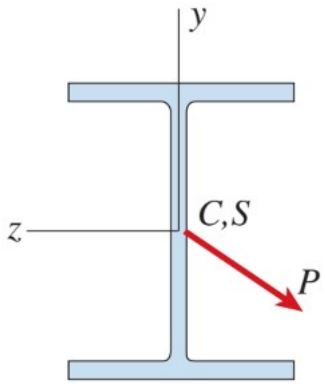
That load can be replaced by the same force acting through the shear center S, plus a torque *T*.

The force through S produces bending. The torque produces torsion.

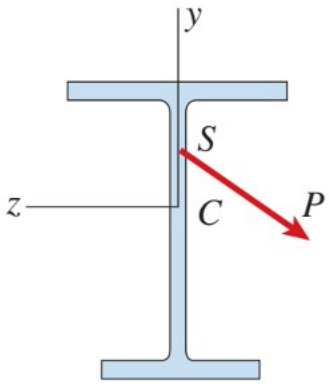


So a lateral load bends without twisting only when it acts through the shear center.

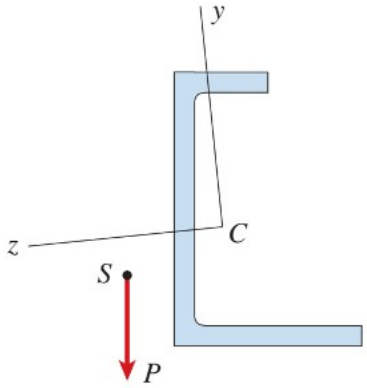
Where the shear center lies



Doubly symmetric: S and the centroid C coincide.



Singly symmetric: both lie on the axis of symmetry.



Unsymmetric: locating S needs more advanced methods.

Thin-walled open sections are a special case, and an important one: they are weak in torsion, so where the load acts matters a great deal.

The two-step procedure

- First, find the shear stresses when the beam bends about one principal axis.
- Then find the resultant of those stresses.
- The shear center lies on the line of action of that resultant.
- Repeat for the other axis. The two lines of action meet at the shear center.

Only singly symmetric and unsymmetric sections need this.

Centerline dimensionsline dimensions

As before, the derivations and calculations use centerline dimensions.

For a channel, *h* is the height between flange centerlines, not the overall depth.

That is satisfactory when the wall is thin.

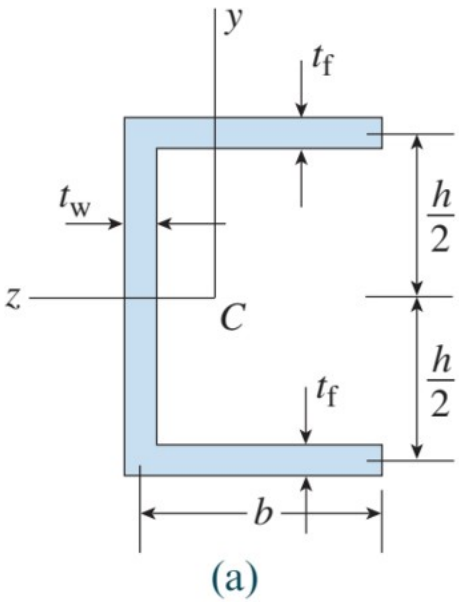
It keeps the arithmetic simple.

The channel section

A channel is singly symmetric, so its shear center lies on the axis of symmetry, the z axis.

To find where on that axis, bend the beam about the z axis and find the line of action of the resultant shear force V_y.

The shear center is where that line meets the z axis.

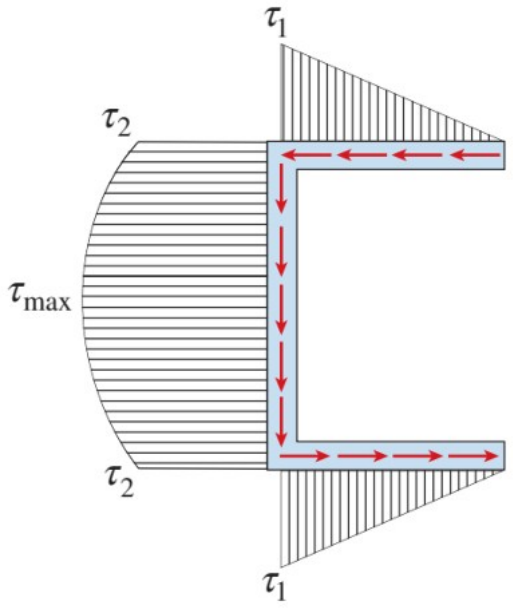


The stresses in a channel

The shear stresses vary linearly along the flanges and parabolically down the web.

Three values are needed to find the resultant:

- τ_1 at the flange-to-web junction
- τ_2 at the top of the web
- τ_{max} at the neutral axis



The stress in the flange

The first moment of the flange about the neutral axis is

$$Q_z = b\, t_f \frac{h}{2} \tag{14}$$

so the stress at the junction, where the width is the flange thickness, is

$$\tau_1 = \frac{V_y\, Q_z}{I_z\, t_f} = \frac{b\, h\, V_y}{2\, I_z} \tag{15}$$

b is the flange width, t_f the flange thickness, and h the height between the flange centerlines.

The stress at the top of the web

The same first moment, but the width at the cut is now the web thickness:

$$\tau_2 = \frac{V_y\, Q_z}{I_z\, t_w} = \frac{b\, t_f\, h\, V_y}{2\, t_w\, I_z} \tag{16}$$

The stress jumps at the junction, because the width jumps from the flange thickness to the web thickness.

The force carried is the same on both sides of the jump; only the area it is spread over changes.

The maximum, at the neutral axis

At the neutral axis the area above the cut is the whole flange plus half the web:

$$Q_z = b\,t_f\,\frac{h}{2} + \frac{h\,t_w}{2}\cdot\frac{h}{4}$$

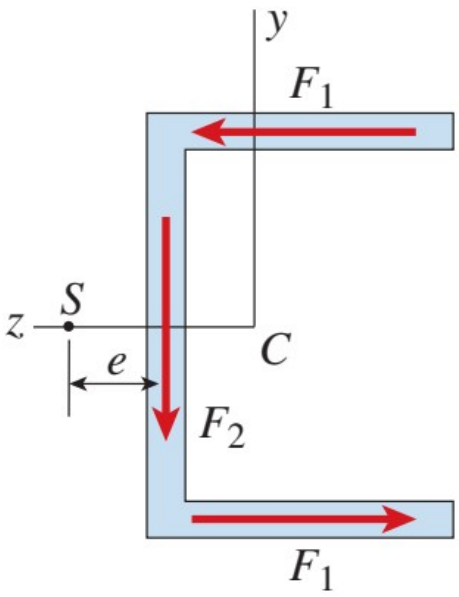
Dividing by the web thickness gives the largest stress in the section:

$$\tau_{max} = \left(\frac{b\,t_f}{t_w} + \frac{h}{4}\right)\frac{h\,V_y}{2\,I_z} \qquad (17)$$

The forces on the section

Each flange carries a triangular stress diagram.

The force is that triangle's area times the flange thickness:



$$F_1 = \frac{\tau_1 \, b}{2} \, t_f = \frac{h \, b^2 \, t_f \, V_y}{4 \, I_z} \quad (18)$$

The two flange forces are equal and opposite in the horizontal direction, so they contribute nothing to the vertical resultant.

Check: the web carries all of it

The vertical force in the web must equal V_y, since the flange forces have no vertical component.

The second moment of the channel, on centerline dimensions, is

$$I_z = \frac{t_w\, h^3}{12} + \frac{b\, h^2\, t_f}{2} \tag{19}$$

The web stress diagram is a rectangle of height τ_2 plus a parabolic segment, so the web force is

$$F_2 = \tau_2\, h\, t_w + \frac{2}{3}(\tau_{max} - \tau_2) h\, t_w \tag{20}$$

Check: F_2 comes out as V_y

Putting that second moment into the expression for the web force:

$$F_2 = \left(\frac{t_w\, h^3}{12} + \frac{b\, h^2\, t_f}{2}\right)\frac{V_y}{I_z} = V_y \qquad (21)$$

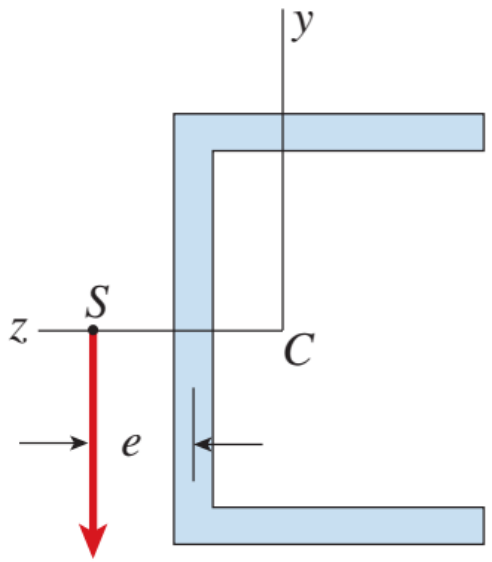
So the web does carry the whole shear force, as expected.

The three forces acting on the cross section are therefore F_1 in each flange and $F_2 = V_y$ in the web.

Locating the shear center

The resultant of the three forces is V_y, and it meets the z axis at the shear center S.

Take moments about S itself. The three forces give $F_1\,h$, and the resultant gives nothing:



$$F_1\,h = F_2\,e \qquad (22)$$

The distance to the shear center

Solve that for e and put in the two forces:

$$e = \frac{F_1\, h}{F_2} = \frac{b^2\, h^2\, t_f}{4\, I_z} \tag{23}$$

and with the second moment of the channel,

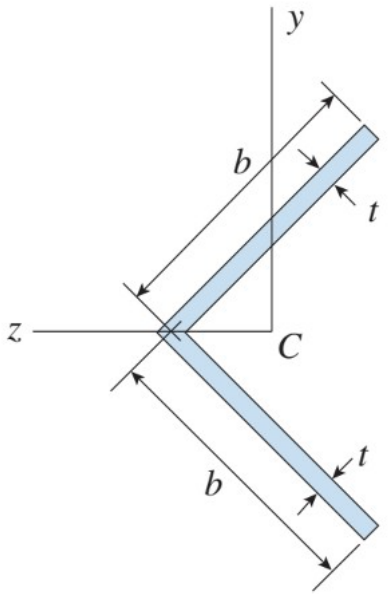
$$e = \frac{3\, b^2\, t_f}{h\, t_w + 6\, b\, t_f} \tag{24}$$

What the answer means

- e is measured from the centerline of the web.
- The shear center lies outside the channel.
- A load through S bends the channel without twisting it.
- A load in the plane of the web adds a twisting couple.
- A load along z gives simple bending about y.

The angle section

Now take an equal-leg angle: each leg is of length b and thickness t.



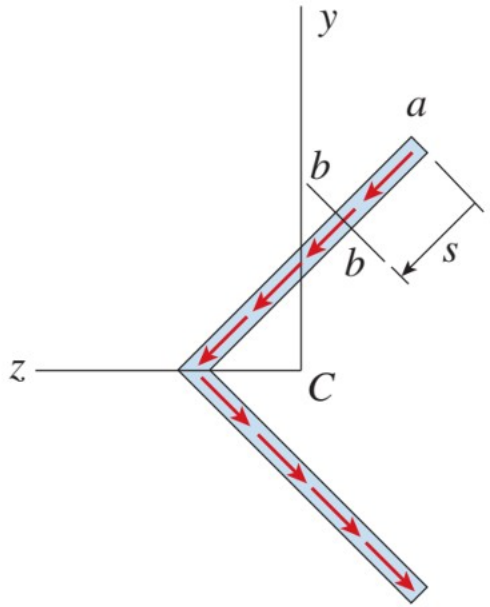
The z axis is an axis of symmetry, and the origin is at the centroid C, so y and z are principal centroidal axes.

The procedure is the same as for the channel, but the result is much simpler.

The first moment along the leg

Measure s along a leg from the free edge a.

The area is s t, and its centroid is (b − s/2)/√2 from the axis:



$$Q_z = s\,t\left(\frac{b - s/2}{\sqrt{2}}\right) \qquad (25)$$

The √2 comes from measuring the distance perpendicular to the neutral axis, while s runs along the inclined leg.

The stress along the leg

Divide by the thickness and multiply by V_y over I_z:

$$\tau = \frac{V_y\, Q_z}{I_z\, t} = \frac{V_y\, s}{I_z \sqrt{2}} \left(b - \frac{s}{2} \right) \tag{26}$$

The second moment of an equal-leg angle about the axis of symmetry is

$$I_z = \frac{t\, b^3}{3} \tag{27}$$

The stress varies quadratically with s.

The stress, and the force in a leg

Putting the second moment in gives the stress at any point:

$$\tau = \frac{3\,V_y\,s}{b^3\,t\sqrt{2}}\left(b - \frac{s}{2}\right) \tag{28}$$

It is largest at the junction of the legs, where s = *b*:

$$\tau_{max} = \frac{3\,V_y}{2\,b\,t\sqrt{2}} \tag{29}$$

and the force in each leg is the area of the parabolic diagram times *t*, which comes to V_y over √2.

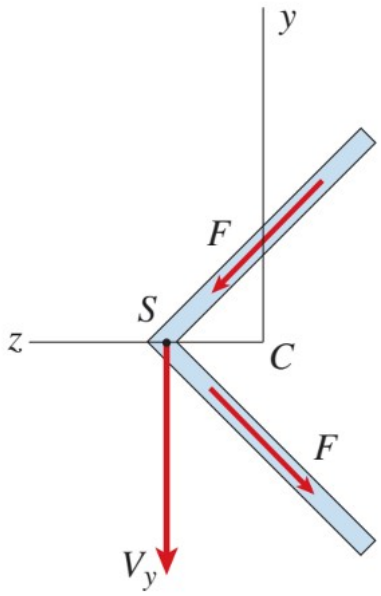
Where the shear center of an angle is

The stresses run along the legs, so each leg carries a force F along it.

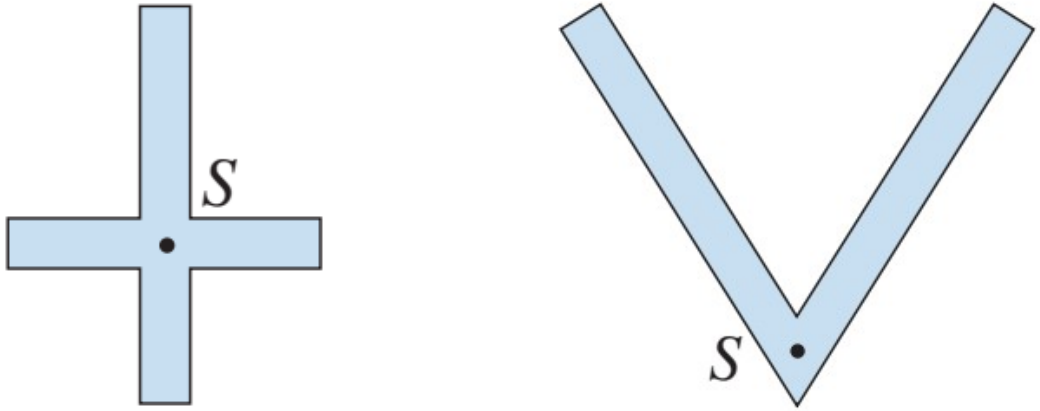
The horizontal components cancel; the vertical ones add to V_y.

The two forces meet at the junction, so the resultant passes through it.

So the shear center is at the junction of the legs.



Two intersecting narrow rectangles



That argument does not need the legs to be equal, or at a particular angle.

In any section of two thin rectangles that meet, each carries a resultant along itself, and the two meet at the junction.

So the shear center is at that junction.

Lecture 7

Summary

The formula for thin-walled open sections

$$\tau = \frac{V_y\, Q_z}{I_z\, t}, \qquad f = \tau\, t = \frac{V_y\, Q_z}{I_z}$$

- s is measured along the centerline from the free edge of the section.
- Q_z is the first moment of the area between the free edge and the cut.
- *t* is the wall thickness at the cut, which may vary.
- The shear force must act through the shear center.

Formula summary: the flange

$$\text{In the flange: } \tau_1 = \frac{b\,h\,P}{4\,I_z}$$

- The flange stress is zero at the free edge and grows with s.
- At the junction with the web it reaches τ_1, its largest value.
- The shear flow there is that stress times the flange thickness.

Formula summary: shear centers

Channel: $e = \dfrac{3\,b^2\,t_f}{h\,t_w + 6\,b\,t_f}$

Equal – leg angle: at the junction of the legs

Two intersecting rectangles: at the junction

Where each result comes from

- The thin-walled formula comes from horizontal equilibrium on an element.
- The wide-flange section applies it to the flange.
- A shear center is where the resultant of the shear stresses acts.
- The channel needs calculation; the angle can be reasoned out.

What this lecture adds

- Lecture 6 covered the web of a wide-flange beam; this adds the flanges.
- The flanges carry shear too, and the flow round the section gives the directions.
- A load must pass through the shear center for the beam to bend without twisting.
- A channel's shear center lies outside the section.

The End of Lecture 7

Next: Torsional shear stress